

Developer Trends Evolution 2024

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Outline

Introduction

Report objectives,
executive summary of
core findings,
and methodology

Tech Stack Trends

In-depth analysis of
languages,
databases, web
frameworks, and
platforms

Developer Profile

Demographics,
education levels, and
future technology
desires



Executive Summary

Strategic Key Point

Modern developers are optimizing for **Accessibility & Ease of Learning (Python)** and **Data Reliability (PostgreSQL)** while rapidly adopting **Cloud Services** into their workflows.

- **PostgreSQL Dominance:** Now the gold standard, used by 49% of developers globally.
- **React Mastery:** React remains the most used Web Framework with the other React based Frameworks.
- **The Rust Phenomenon:** Continues as the most "Want To Learn" language.
- **Cloud Leadership:** AWS holds a commanding lead, followed by Azure and GCP.



Introduction

- **Report Objectives**
To synthesize data from over 65,000 developers to identify emerging trends and strategic shifts in technology adoption.
- **Target Audience**
Designed for CTOs, Engineering Managers, and Product Leads seeking data-driven decision support for their technical roadmaps.
- **Strategic Value**
Providing essential context for hiring, training, and tech-stack selection for the 2024-2025 cycle.



Methodology

- **Data Source**

Stack Overflow Developer Survey 2024, collected from 65,000 developers about the tools they use, the technologies they want to learn, their experiences at work, and much more.

- **Data Cleaning & Filtering**

Removed incomplete or insincere responses to ensure dataset integrity.
Filtered out extreme outliers using statistical standards.

- **Feature Engineering & Categorization**

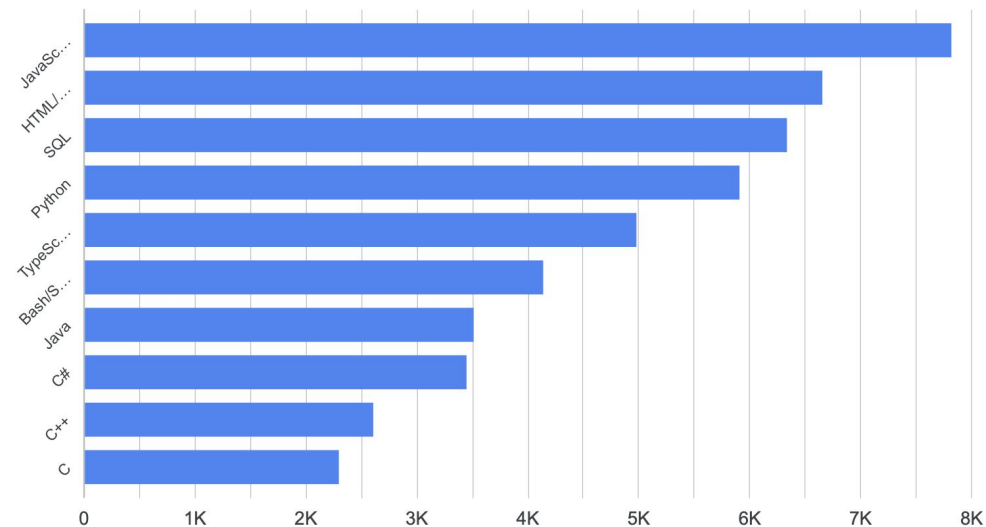
Parsed and standardized complex multi-select survey questions into distinct, analyzable variables.



Programming Languages Trends

Top 10 Languages Used

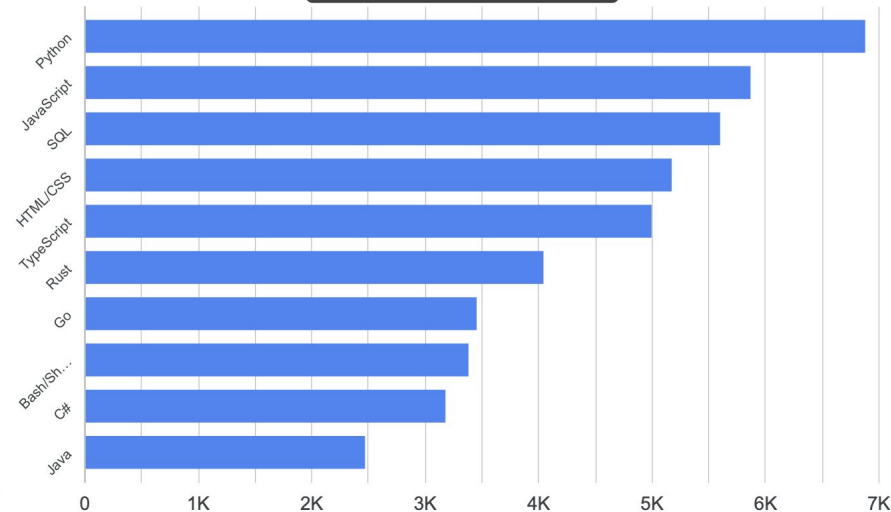
Record Count



Top 10 Languages Desired Next Year

Record Count

Top 10 Languages Desired Next Year





Programming Languages Trends :

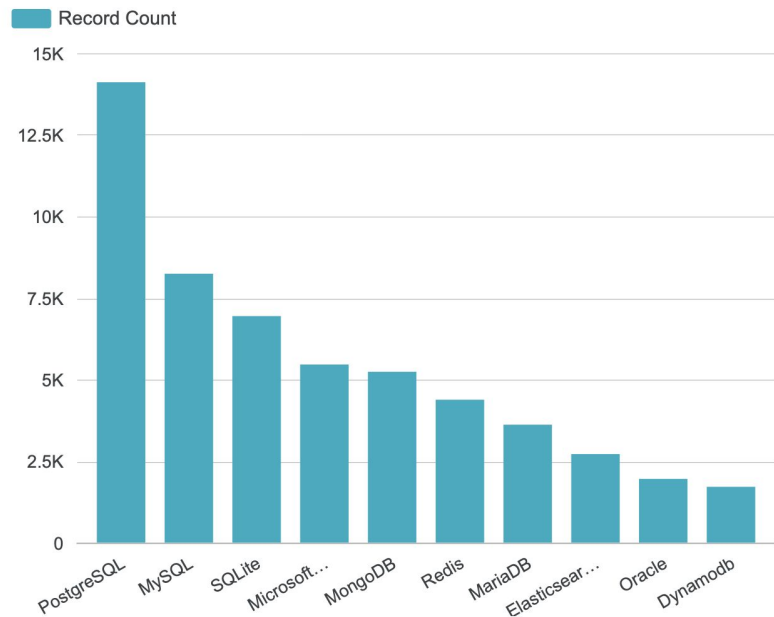
Key Points

- **Language Shift (2024 → 2025):** Python rising to share the #1 spot with JavaScript, overtaking HTML/CSS.
- **SQL Dominance:** Remains the undisputed top language for database queries.
- **Rust > C++:** Outranking C++ as the new preferred low-level programming language.
- **Momentum in Go:** Emerging rapidly as a key language for modern development.

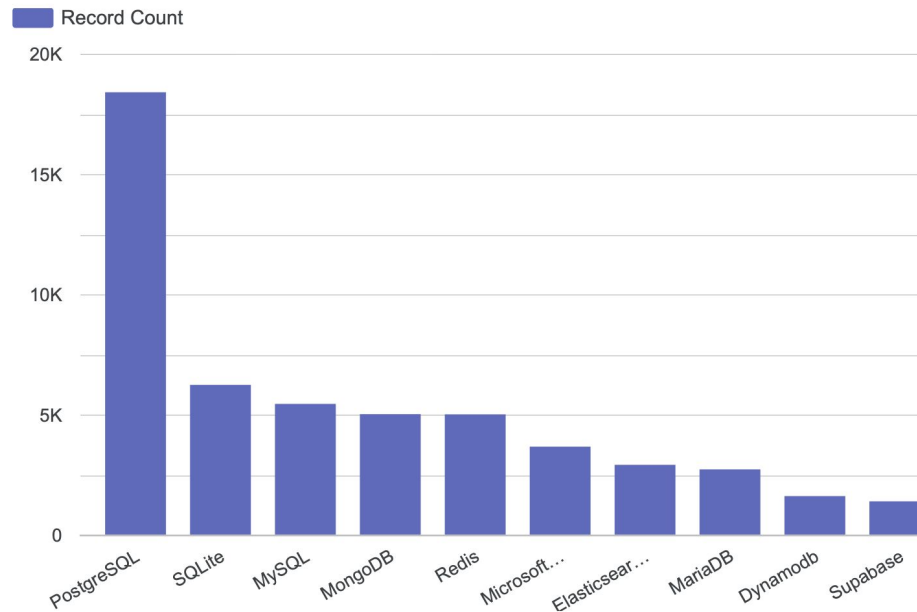


Database Trends

Top10 Databases Used



Top 10 Databases Desired Next Year

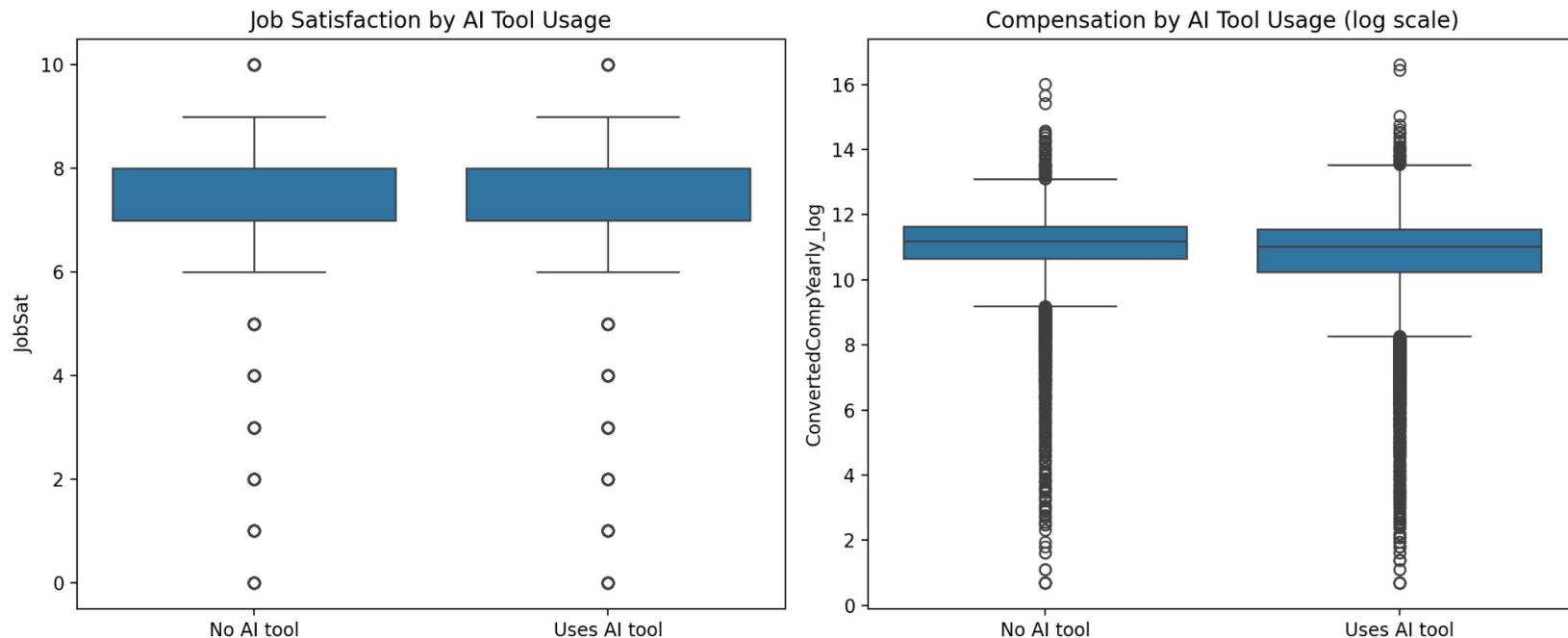




Database Trends : Key Points

- **Unrivaled PostgreSQL:** Continues to hold the top spot, proving its reliability as the industry standard.
- **Strength of Relational DBs:** SQL-based databases show enduring dominance and are projected to stay on top through 2025.
- **MongoDB as the NoSQL Choice:** Retains its position as the most preferred NoSQL database among developers.
- **Redis Climbs the Ranks:** Leaping from 6th to 4th place, reflecting the rising demand for fast, queue-based solutions to power real-time services like app booking systems.
- **The Rise of Supabase:** A notable new entry gaining rapid traction in the modern database ecosystem.

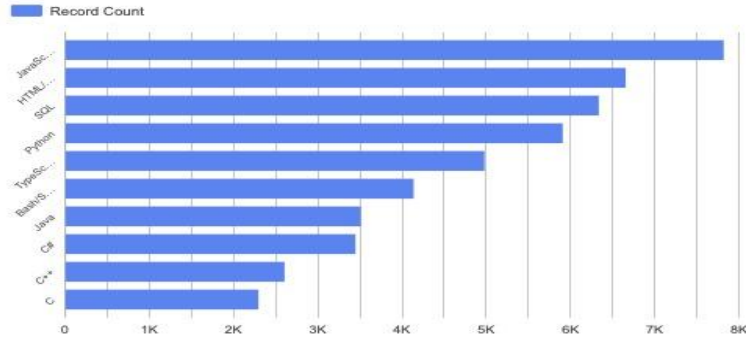
AI Tool Adoption & Developer Outcomes



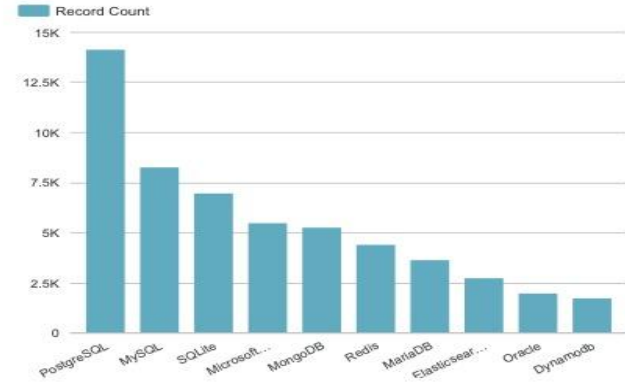
Median JobSat and compensation are nearly identical between groups — a statistically significant difference ($p = 0.0016$) with limited practical impact at this sample size.

Dashboard : Current Technology Usage

Top 10 Languages Used



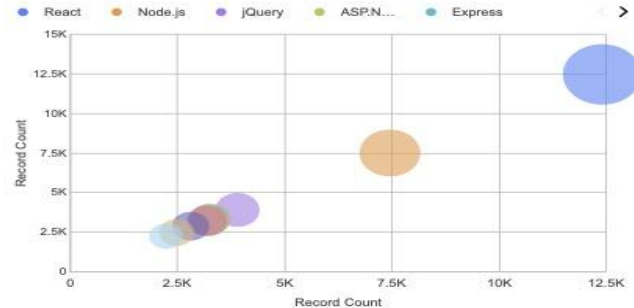
Top 10 Databases Used



Top 10 Platforms Used

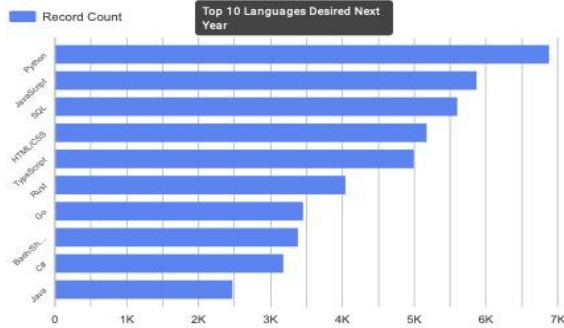


Top 10 Web Frameworks Used

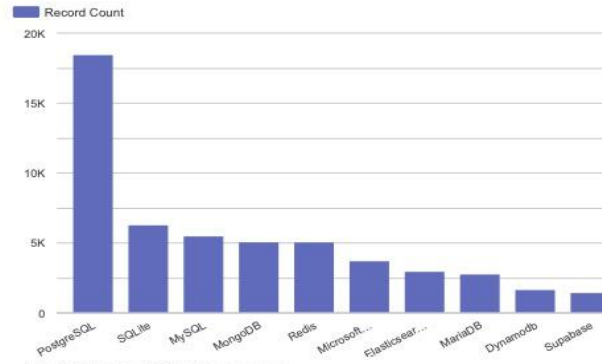


Dashboard : Future Technology Trends

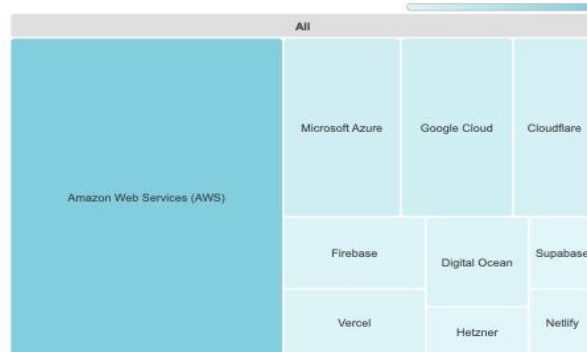
Top 10 Languages Desired Next Year



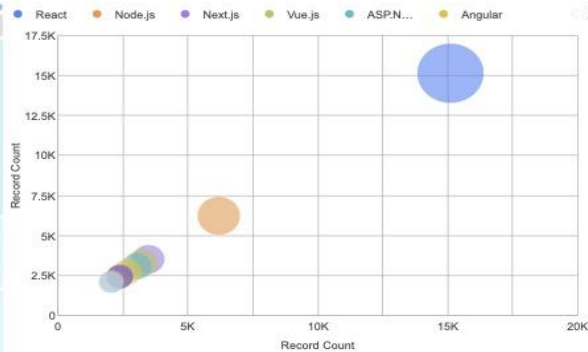
Top 10 Databases Desired Next Year



Top 10 Desired Platforms



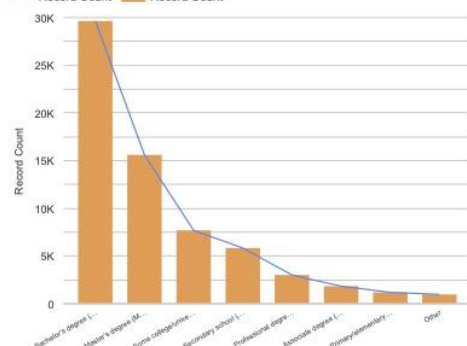
Top 10 Desired Web Frameworks



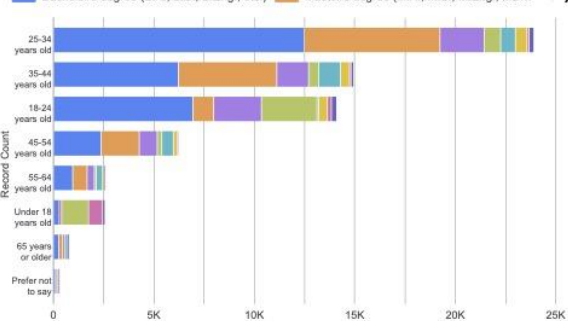
Respondents by Age



— Record Count ■ Record Count



 Bachelor's degree (B.A., B.S., B.Eng., etc.)
  Master's degree (M.A., M.S., M.Eng., MB...





Dashboard insights : Tech Stack Evolution

- **Language Shifts:** TypeScript accelerates while traditional HTML/CSS reliance softens; Java shows a continuous decline.
- **Database Balance:** Relational DBs dominate, but specialized NoSQL (MongoDB, Redis, DynamoDB) remains highly desirable.
- **Framework Longevity:** React-based web frameworks secure their position for the foreseeable future.



Dashboard insights : Infrastructure & Demographics

- **Cloud as Baseline:** AWS leads, but cloud platforms are broadly becoming universal, multi-vendor infrastructure.
- **Developer Profile:** Core workforce aged 20–40, predominantly holding Bachelor's or Master's degrees.
- **Geographic Shift:** US-centric data reveals more tech job opportunities on the East Coast than the traditional West Coast.



Overall Findings & Implications

Key Findings:

1. Python (for ease and speed) and TypeScript have become the go-to languages, while PostgreSQL remains the absolute bedrock for data.
2. Cloud services (AWS, Azure) are no longer competitive advantages; they are universal baselines.
3. AI tool adoption is now widespread among developers, but shows no meaningful link to reported satisfaction or pay — suggesting it's becoming a baseline expectation rather than a differentiator.

Implications:

1. Build for agility and reliability by pairing highly accessible languages (Python) with battle-tested databases (PostgreSQL/Redis).
2. Architect systems to be flexible and cloud-agnostic, rather than locked into a single provider.
3. Don't treat AI tool as a differentiator in hiring yet— focus instead on what candidates can do with those tools, since usage alone doesn't correlate with better outcomes.



Conclusion : Key Takeaway

- **Future-Proofing the Stack:** The shift toward Python, TypeScript, and Rust highlights a market move toward performance, safety, and scalability.
- **Adopt Early:** Invest in the React/TypeScript ecosystem and evaluate Rust for performance-critical systems.
- **Optimize Data:** Standardize on PostgreSQL for data integrity, but integrate Redis/MongoDB to handle specialized, real-time app demands.
- **Infrastructure Readiness:** Prepare for a multi-cloud future as platforms become a basic infrastructure standard.



Slide Reference

<https://stackoverflow.blog/2024/08/06/2024-developer-survey/>